

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

PRACTICAL - 8

AIM :- To implement the Naive Bayes learning algorithm for predicting whether a customer will purchase a product based on their age and estimated salary.

Dataset: customer_purchase.csv

Age,EstimatedSalary,Purchased

19,19000,0
35,20000,0
26,43000,0
27,57000,0
19,76000,0
27,84000,0
26,80000,0
32,150000,1
28,100000,1
30,110000,1
35,50000,0
40,70000,1
45,80000,1
48,90000,1
50,100000,1
22,25000,0
25,30000,0
29,45000,0
31,60000,1
38,65000,1
42,75000,1
46,85000,1
52,120000,1
55,130000,1
60,140000,1
21,18000,0
24,35000,0
33,45000,0
36,55000,1
39,95000,1

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Code:-

```
#T101 RAJDEEP M PARAB
# AIM: Implement Naive Bayes learning algorithm
# for customer purchase prediction.
import pandas
from sklearn import model_selection
from sklearn.naive_bayes import GaussianNB
# Read local customer dataset
dataframe = pandas.read_csv("customer_purchase.csv")
# Display first five records
print(dataframe.head())
# Convert dataset into array
array = dataframe.values
# Input features
X = array[:, 0:2]
# Output class
Y = array[:, 2]
# Random seed
seed = 7
# Create Naive Bayes model
model = GaussianNB()
# Evaluate model using cross-validation
results = model_selection.cross_val_score(
    model, X, Y, cv=10
)
# Display accuracy
print("Accuracy:", results.mean())
```

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Output:

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
- RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-8/Practical-8.py
  Age  EstimatedSalary  Purchased
0  19          19000         0
1  35          20000         0
2  26          43000         0
3  27          57000         0
4  19          76000         0
Accuracy: 0.8999999999999999
>>>
```

